IS222 – Database Management Systems

Semester I, 2023

Mode: Face to Face & Online (Only Online Submission Required)

Assignment 1 - Category: Individual Assignment

Due Date: [3rd April 2023]

Weight: 15%

Assignments Objectives

- This assignment tests the following course learning outcomes and the associated CBOK.

Attributes:

CLO 2: Construct a data model for a relational database management system.

1. Determine the business rules to describe the data required for a given business process.
2. Identify the entities and attributes for a relational database using the business rules.
3. Draw the proposed database's conceptual model (ERD) using MS Visio.
4. Create the Data Dictionary for the proposed database.
5. Implement the proposed database using MS Access 2016.
6. Build simple queries, forms, and reports for your database using MS Access 2016 or any relevant.

***Note:** The attached marking rubric on pages 5-7 will be used to assess this assignment.

Prerequisites

Students must complete this assignment independently, individually, and **NOT** in groups.

Before you commence this assignment, you should have at least completed the following:

- Reading chapters 1-4 of the textbook.
- Completed all required Moodle (Course) activities from Weeks 1 to 5.
- It would be best if you covered all materials and lab exercises on designing databases with MS- Access and MS Visio 2016.

Assignment Overview

Scenario:

The Tulsi's Spare Parts & Logging Industry

The Tulsi's Spare Parts & Logging Industry includes retailers of new and used car parts and accessories, auto mechanics, and installers of auto accessories located in Ba, Fiji. Mr. Tusi Ram plans to develop a database to store data regarding its customers, motor parts, and staff. The company has 12 branches in different towns/cities around Fiji. There are 12 branches of the business located throughout Fiji's towns and cities. The company wants to eliminate the manual file system due to possible data loss, company records are not adequately organized, etc. To keep track of information about his clients, staff, and motor parts, Mr. Tusi Ram intends to create a database. Due to potential data loss, improper organization of company documents, etc., the company wants to eliminate the manual file system. When a customer arrives at the shop, the friendly staff greets the customer and identifies whether the customer is a new customer, or the customer has bought some parts previously. If the customer is new, the details of the customer is noted and stored in the database. The details about the customer that are kept by the company include the Customer Number (unique for each customer), the customer's name, residential and postal addresses, and home and work phone numbers. A customer can be classified as private or business. Business customers buy most parts from the company so they are eligible for some discount. Private customers can also get discount but it will be less than the business customer.

The staff then serves the customer by showing the parts required and gives valuable suggestions to customers as well. The company stores the part number, part description, part price, quantity on hand, quantity on order, etc. When a customer buys part(s), an invoice is issued. For each invoice, the invoice number, invoice date, customer number, customer name,

part number(s), part description(s), quantity, price, staff number and name who sold the parts are recorded. For each part, the supplier information, quantity on hand and branch location should also be stored in the database. According to company's policy, a supplier can supply a maximum of 5 different parts only.

Staff can generate many invoices. For each staff, staff number (unique for each staff), name, office number, office name, branch location, branch phone, etc is recorded.

As part of this assignment, you will create an MS Word document, an MS Visio file, and an MS Access database. Only the MS Access database and MS Word document should be submitted. The submission instructions are provided later.

Requirements:

Part A. Create an MS Word document showing the following:

1. Business Rules that you can identify for the Tulsi's Spare Parts & Logging Industry Company.
2. Entity Relationship Diagram (ERD) for the Tulsi's Spare Parts & Logging Industry Company.

Note:

- Create the ERD in Visio or other software, and copy and paste it into your MS Word document.
- There is no need for you to submit the MS Visio file.

3. Data Dictionary for your database

Part B. Create an MS Access Database

1. Using your ERD and Data Dictionary, create the required tables in an MS Access Database.

2. Enter some sample data into the tables, for example, 5 – 20 records per table.

3. Create queries to answer the following questions:

   a) List all the invoices that have more than one part listed in them. The list should be sorted by invoice number.

   b) List the staff number and name that has sold more than two parts. The list should be sorted by last name.

c) Create a parameterized query, which takes in the supplier name as a parameter and lists all the parts supplied by a particular supplier.

Submission Guidelines:

- Upload the MS Access file and MS Word file using Moodle. The name of the files should be your student ID, for example, S11082321.docx, etc.
- Incorrect submissions will result in a loss of marks or a zero mark.
- LATE SUBMISSION = ZERO (0)
    - Incorrect submissions will result in loss of marks or simply a grade of zero being awarded.
    - Late Submission = Grade of Zero (0) for Assignment 1.

Plagiarism

No two students should submit the same assignment. Any such case will be dealt with severely, with actions including awarding a mark of zero, forwarding the case to the Head of School, and reporting the matter to the Student Disciplinary Committee.

Queries

- Maintain contact through Moodle discussion forum named Assignment 1 Discussion Forum
- Feel free to make an appointment via email with the Course Coordinator for any urgent matters regarding Assignment 1.

Assignment 1 Assessment Rubric

IS222 – Database Management Systems – S1)23					
Assignment 1 Marking Rubric					
Marker:					
Student ID:					
Total Mark:	0	/78			
Question	Details				
Partts:	Database Design Report – CBOK		Unsatisfactory	Satisfactory Good	Marks (x/61)
	Business Rules 2 marks each for BR			14 Marks	

	For each business rule: • Appropriate and relevant business rules have been identified that capture the business problem. • Business rules are clear, concise, and stated in the correct format.				
	ERD 2 marks each for each entity • 0.5 mark correct entity name (for each entity) • 0.5 mark for proper attribute name (for each entity) • 0.5 mark for the required attribute (for each entity) • 0.5 mark for PK/FKs (for each entity) 1 mark for each relationship = 9 marks • 0.5 mark for strength • 0.5 mark for cardinality/Connectivity	37 Marks			

		12 Marks

	Data dictionary 2 marks for each section Identified below! • Marks should be given for: • Attribute names eg. CUS_LNAME • Data types -- phone cannot have numbers • Attribute descriptions • Required fields • Format - TEXT/Char • PK/FK and FK Referenced Table				
	Queries 5 marks for each query:	15 Marks			
	Query 1: Sorting and required information to be displayed (5 marks) **Query 2:** Sorting and required information to be displayed (5 marks) **Query 3:** Parameterized and required information to be displayed (5 marks)				

General Comments

CBOK	Criteria	
A. Data and Information Management (Milestone)	Unsatisfactory I. Incomplete/incorrect ERD II. Incorrect tables, connections, cardinalities etc. III. Inappropriate tools used for table design IV. Mostly incorrect SQL queries V. Data is not stored and managed using inappropriate tools.	
	Satisfactory	

	I. Correct ERD diagram II. Appropriate tables/entities have been identified (or defined) with appropriate fields III. Design of table and use of DBMS IV. Correct SQL query for a given question V. data is the store and managed using a standard tool	
	Good	
	I. All satisfactory and II. Exceptional ERD. All possible scenarios are considered.	
B.Knowledge Area: ICT Problem-Solving Abstraction (Benchmark)	Unsatisfactory	
	I. Incorrect flowchart or UML/ERD diagrams. II. Inappropriate tools used for UML/ERD diagrams, such as MS Word. III. Poor architecture and design make future updates hard to incorporate.	
	Satisfactory-Good	
	I. all relevant constraints and requirements are mentioned II. Correct flowchart or UML/ERD diagrams using an appropriate tool III. Appropriate choice of the architecture and design	
C.Understanding of ICT Profession (Benchmark)	Unsatisfactory	
	I. Unable to analyze the recent issues whilst solving a specific ICT problem.	
	Satisfactory	
	I. Poor judgment on the analysis of the recent issues whilst solving a specific ICT problem.	
	Good	

	I. Analyze the recent issues with proper justification and logical arguments whilst solving a specific ICT problem.

The End.